VERA — Vision-Enabled Reasoning Agent — System Architecture

v1.88.0 — Auto-calibrated A/V sync delay for spectrum display

Optional Matter processes (systemd user services, Node.js 24)
vera-matter-controller
OHF matter-server
local dashboard/WS 127.0.0.1:5580
Google Home Matter controller
Matter / IPv6 LAN
independent Matter fabric
Matter / IPv6 LAN
Tapo P316M / Matter Wi-Fi devices
native multi-admin sharing
vera-matter-bridge
src/matter/bridge.mjs
VERA Audio Mute, UDP 5541
~/.local/share/vera-matter
separate controller + bridge fabrics
private persistent credentials
Root-owned deployment controller (separate from VERA)
vera update + authenticated System GUI
/run/vera-updater/control.sock
peer-checked fixed operations
vera-updater.timer
vera-updater-check.service
local-time installation window
local Unix socket
vera-updater.service
/opt/vera-updater/src/updater
stage / deploy / verify / recover
ZeroMQ IPC
verified full-duty kernel PWM
before thermal restart
manifest + verified assets
atomic journal
deployment.readiness RPC
loaded version/revision/PID
fresh telemetry + service/hardware health
NF-A6x25 PWM Fan
PWM ← GPIO13 (kernel HW PWM, 25 kHz)
TACH → GPIO6 (2 ppr)
GitHub stable releases
mazum01/desktop-assistant
HTTPS, exact tag/commit
/var/lib/vera-updater
protected backups + transaction state
post-reboot continuation
tach
PWM
Hardware (Raspberry Pi 5)
TMP102
(VERA, I²C 0x48)
Pi Cam Module 3 Wide
(CSI-0)
TMP117
(VERA3, I²C 0x48)
Hailo AI Accelerator
(Hailo-8 / Hailo-10H, PCIe)
Pan servo + optional tilt
PCA9685 configurable channels
180deg direct-drive / legacy DS3218
configured safe travel limits
reSpeaker XVF3800
(USB mic array)
reSpeaker XVF3800 output →
attached speaker
Waveshare ESP32-S3-Touch-LCD-1.47
172×320 SPI TFT (touch disabled)
(BLE mouth/status display + BLE OTA)
C6 touch + legacy profiles supported
Display microSD (S3)
/videos/*.mjpg or *.mjpeg
baseline JPEG streams, no audio
Optional QMI8658 (identity-probed)
not specified on S3 hardware
roll/pitch/relative yaw, no compass
CSI
I²C
PCIe control
(temp only)
PCIe
I²C/PWM
USB
SPI + GPIO
SD_MMC 1-bit
CLK16/CMD15/D0=17
I2C 0x6A/0x6B
identity + checked samples
Drivers (src/)
vision/camera.py
(picamera2)
thermal/hailo_temp.py
(Hailo die temp via HailoRT control,
monitor only)
thermal/tmp117.py
audio/version_announcer.py
perception/hailo_inference.py
(VStreams + InferModel API)
perception/hailo_models.py
config/hailo/models.yaml
probe / select / download HEFs
python3 -m ...hailo_models --install
motion/servo_controller.py
motion/visual_servo.py (optional controller)
stable target lock + damped visual feedback
identified camera/servo Jacobian
config/tracking_calibration.json (local)
core/body.py
config/body.yaml (machine-local identity/tuning)
config/body_profiles.yaml (hardware/controller)
body-tagged runtime servo geometry
motion/head_tracker.py (original body)
legacy Kalman + minimum-jerk pan tracking
new-body idle motion controller
motion/gaze_tracker.py (new body default)
world-bearing Kalman + minimum-jerk pan/tilt
smooth pursuit · fixation hold · saccadic suppression
audio/input.py
TMP117 ID
matched SCRFD HEF
thermal/fan_tach.py
(lgpio edge cb)
thermal/fan.py
(sysfs HW PWM
+ lgpio fallback)
thermal/temperature_sensor.py
(probes TMP117 / TMP102)
audio/tts.py
(espeak-ng)
perception/object_detector.py
(YOLOv11l on-chip NMS; YOLO26m+ONNX H8 fallback)
perception/face_detector.py
(SCRFD decode + Haar fallback)
latest_frame()
latest_frame()
renders to
thermal/thermal_manager.py
audio/output.py
matched ArcFace HEF
wiring, range, limits, tilt capability
controller and mount direction
matched depth HEF
BLE GATT (OTA)
External processes
scripts/desktop-assistant
CLI: ping / status / pan / say / watch / ...
display video list/play/stop/status
waits for correlated device reply
~/.local/share/desktop-assistant/
telemetry.db (SQLite ring)
faces.db (identity registry)
thumbs/ (64×64 JPEG face crops)
3rd-party subscribers
(dashboards, future ML pipelines)
Web Browser
http://<pi>:8080
firmware/upload_esp32_display_ble.sh
+ nimbleota.py uploader
(BLE firmware push, no USB)
ZMQ REQ/SUB
writes telemetry.db
ZMQ SUB
loopback GET/PUT /api/audio/mute
HTTP/WS :8080
latest_frame()
latest_observation()
latest_frame()
writes faces.db
ZMQ PUB→SUB
(cross-process)
vision.mono_depth_map
strict music.get_muted / set_muted RPC
IPCClient RPC
(music/podcast request/actions)
ZMQ PUB→SUB
(cross-process)
VisionService / RawCameraService
→ vision.frame_ready / vision.frame2_ready
← vision.capture_still
RPC vision.get_focus / set_focus
RPC camera2.get_focus / set_focus
Web: GET|PUT /api/settings/camera{,2}/focus
per-device persisted AF / manual diopters
+ latest_frame()
TelemetryService
← thermal.* / motion.position / audio.level
→ telemetry.flush
SQLite ring buffer
IPCBridge (core)
PUB ipc:///tmp/desktop-assistant.pub
REP ipc:///tmp/desktop-assistant.rep
SUB ← thermal.pub (upstream)
cmds: ping / topics / last / publish / status
AVService
← av.say / beep / utterance / announce_version
→ av.spoke / av.speaking_started / version_announced
ObjectService
← vision.frame_ready / object.set_enabled
← perception.faces (selective label association)
→ perception.objects (held overlays + fresh persons)
→ object.enabled_changed (startup + changes)
2fps · overlay max_objects cap · BGR→RGB fix
WebService (FastAPI)
GET / — dashboard SPA
GET /stream — MJPEG 640×360 feed
WS /ws — live JSON status
GET /api/faces — list (has_thumb flag)
GET /api/skills — registered skills
GET /api/faces/{id}/thumb — face JPEG
PUT /api/faces/{id} — rename
DEL /api/faces/{id} — delete one
DEL /api/faces — delete all
GET|PUT /api/settings/quiet-hours
GET|PUT /api/settings/audio
GET|PUT /api/settings/voice
GET|PUT /api/settings/servo/channels
PUT /api/settings/servo/tilt-limits
GET|PUT /api/room
GET|PUT /api/music/* — proxied via IPCClient
GET|POST /api/podcasts/* — proxied via IPCClient
POST /api/say|pan|tilt|version|utterance
port 8080
ThermalService
— thermal.temp
— thermal.fan
— thermal.rpm
— thermal.critical
— thermal.error
— thermal.hailo
PerceptionService
← vision.frame_ready
→ perception.faces / perception.error
+ FaceEmbedder (ArcFace) + FaceRegistry
re-emit
MotionService
shared PCA9685, independent pan / tilt
50Hz bounded quintic profiles
← motion.pan_to / tilt_to / center / relax / stop
← motion.calibration_lock (expiring lease)
← motion.set_channels / set_limits / set_tilt_limits
→ motion.position / moved / settings / error
→ motion.channels_changed / limits_changed / tilt_limits_changed
RPC motion.get_status / motion.set_channels
TrackingService (body-selected controller)
original: legacy Kalman/minimum-jerk pan only
new: gaze (Kalman bearing) pan/tilt; visual_servo optional
← perception.faces / perception.objects (fresh persons)
← object.enabled_changed (person-seek gate)
← motion.position / settings / limits_changed
Enabled + ready servos required
Recognized / largest face; 1s loss handoff
Largest person + 15% stability margin
→ motion.pan_to / tilt_to (new body only)
← tracking.start_autotune / cancel_autotune (visual_servo)
→ motion.calibration_lock / av.say
← av.spoke (announce before probes)
→ tracking.debug / autotune_progress / autotune_done
Tilt: travel limits and center +/-20deg
independent random idle pan · speaking nod
AudioCaptureService
→ audio.level / chunk
+ latest_chunk()
latest_chunk() + chunk_index()
VoiceCommandService
← audio stream/VAD
→ voice.state / wake / transcript / intent
→ av.utterance (routed transcript)
wake-window + dialog context
IPCBridge (media)
PUB ipc:///tmp/desktop-assistant-media.pub
REP ipc:///tmp/desktop-assistant-media.rep
SUB ← core.pub (upstream)
RPCs: music.get_state/set_volume/...
podcast.search/subscribe/play/...
systemd: desktop-assistant-core.service
RoomService
← room.set
→ room.updated / av.say
persists room name · visual scene
baseline histogram · sustained
divergence detection → TTS prompt
MonoDepthService
scdepthv3 on Hailo-8 / Hailo-10H
→ vision.mono_depth_map
relative 320×256 depth, 3 Hz
Arduino firmware
firmware/vera_display
(ESP32-S3 process boundary; C6 profiles retained)
sd_video + JPEGDEC / orientation fusion
main-loop queue + reserved angle footer
NimBLEOta service (BLE OTA, 0x8018)
BLE GATT
replies / playback events / IMU status
BLE notifications
DisplayService
← system.startup_status / service.started / *.error
← display.spectrum / av.speaking_started / av.spoke
← display.video (list/play/stop/status)
→ display.status / display.error / display.reply
→ display.reply.<request_id> / display.device_event
→ display.spectrum_latency (audio→BLE→ack timing)
BLE command writes (MTU; eq frames write-without-response)
PlaybackSpectrumService
← music.state_changed / podcast.playback
pw-record on DA EQ sink monitor (pre sync-delay) + FFT
→ display.spectrum
← display.spectrum_latency → AvSyncCalibrator
pw-cli set-param EQ sync_delay (A/V sync)
FaceService
← perception.faces / face.meet
→ av.say (greetings)
→ face.identified
cooldown-based re-greet
ClockService
+ ClockAnnouncer
→ av.say (hourly/half-hourly)
respects QuietHours
SkillsService
← av.utterance / thermal.temp / thermal.hailo
→ av.say / vision.describe / av.tell_joke
→ face.meet / motion.pan_to / music.*
→ object.set_enabled / music.set_volume
→ tracking.set_face_tracking
SkillRegistry — first-match dispatch
17 skills: help · system_status · volume
quiet_hours · weather · reminder
smart_home · news + 9 others
ConfigField framework: enabled / config_schema
NotificationService
← thermal.temp / thermal.hailo / perception.faces
← av.speaking_started / av.spoke
→ av.say (thermal, Hailo, absence alerts)
rate-limited · quiet-hours aware
systemd: desktop-assistant-thermal.service
IPCBridge (thermal)
PUB ipc:///tmp/desktop-assistant-thermal.pub
REP ipc:///tmp/desktop-assistant-thermal.rep
MusicService
← music.play / stop / pause / next
← music.thumbs_up / thumbs_down / set_volume
Pandora / audio playback
PodcastService
Apple Podcasts search + RSS subscriptions
← podcast.play/pause/resume/stop/refresh
→ podcast.playback / subscriptions_updated
extends
config R/W
suppresses
greetings
suppresses
announcements
suppresses
alerts
Core (src/core/)
core MessageBus
(in-process pub/sub)
version.py
(reads /VERSION)
thermal MessageBus
(in-process pub/sub)
Service base
(start/tick/stop)
QuietHours
(enabled / start / end
loaded from config/quiet_hours.yaml)
media MessageBus
(in-process pub/sub)